

Pharmacovigilance & Adverse Drug Reaction (ADR) Analysis of Paracetamol

Pharmaceutical Research Internship

Prepared By: Alakuntla Shruthi

1. Introduction

Paracetamol, also known as acetaminophen, is one of the most widely used medicines for relieving pain and reducing fever. It is available over the counter and is commonly prescribed for headaches, muscle pain, toothache, fever, and mild to moderate pain. Because of its effectiveness and relatively good safety profile at recommended doses, Paracetamol is used by millions of people worldwide.

Although Paracetamol is generally considered safe, excessive use or overdose can lead to serious adverse effects, particularly liver damage. Therefore, continuous monitoring of its safety through pharmacovigilance is essential to protect patients and improve healthcare outcomes.

2. What is Pharmacovigilance?

Pharmacovigilance is the science and practice of detecting, assessing, understanding, and preventing adverse effects or any other medicine-related problems. It helps healthcare professionals and regulatory authorities monitor the safety of medicines after they are approved for public use.

The main objectives of pharmacovigilance are:

- Monitor the safety of medicines after marketing.
- Detect new or rare adverse drug reactions.
- Improve patient safety.
- Promote the safe and effective use of medicines.
- Support regulatory decisions regarding drug approval and labeling.

3. Mechanism of Action of Paracetamol

Paracetamol works primarily by reducing the production of prostaglandins in the central nervous system. Prostaglandins are chemicals that contribute to pain and fever.

Mechanism:

- Inhibits cyclooxygenase (COX) enzymes in the brain.
- Reduces the production of prostaglandins.
- Relieves pain (analgesic effect).
- Lowers body temperature (antipyretic effect).

Unlike non-steroidal anti-inflammatory drugs (NSAIDs), Paracetamol has very little anti-inflammatory activity.

4. Reported Adverse Drug Reactions (ADRs)

Although Paracetamol is safe at therapeutic doses, some adverse drug reactions have been reported.

Common ADRs

- Nausea
- Vomiting
- Skin rash
- Dizziness
- Mild allergic reactions

Serious ADRs

- Liver toxicity (hepatotoxicity)
- Acute liver failure due to overdose
- Kidney damage (rare)
- Severe allergic skin reactions such as Stevens–Johnson Syndrome
- Anaphylaxis (very rare)

The risk of serious ADRs increases significantly when Paracetamol is taken above the recommended dose or combined with alcohol.

5. Causes of Adverse Drug Reactions

Several factors can increase the likelihood of ADRs:

- Taking more than the recommended daily dose.

- Long-term excessive use.
- Alcohol consumption while using Paracetamol.
- Existing liver disease.
- Drug interactions with certain medications.
- Accidental overdose from taking multiple products containing Paracetamol.

6. Measures to Reduce Risks

The following precautions can help minimize adverse drug reactions:

- Take only the prescribed or recommended dose.
- Avoid exceeding the maximum daily dosage.
- Read medicine labels carefully before use.
- Avoid alcohol during treatment.
- Consult a healthcare professional before long-term use.
- Store medicines safely away from children.
- Seek immediate medical attention if an overdose is suspected.

7. Importance of Pharmacovigilance

Pharmacovigilance plays a vital role in ensuring medicine safety throughout its lifecycle.

Benefits include:

- Early detection of rare side effects.
- Improved patient safety.
- Better awareness among healthcare professionals.
- Prevention of medication errors.
- Continuous improvement of drug safety information.
- Support for regulatory authorities in updating safety guidelines.

Organizations such as the World Health Organization (WHO), national regulatory agencies, and hospitals collect and analyze ADR reports to identify potential safety concerns.

8. Conclusion

Paracetamol is one of the safest and most effective medicines for pain relief and fever reduction when used according to recommended guidelines. However, misuse, overdose, or prolonged use may result in serious adverse drug reactions, particularly liver damage.

Pharmacovigilance helps identify, evaluate, and prevent these adverse effects through continuous monitoring and reporting. Proper patient education, adherence to dosage recommendations, and timely reporting of adverse reactions contribute to safer use of Paracetamol and improved public health.

References

1. World Health Organization (WHO). Pharmacovigilance Guidelines.
2. U.S. Food and Drug Administration (FDA). Acetaminophen Safety Information.
3. DrugBank Online – Paracetamol Drug Information.
4. National Center for Biotechnology Information (NCBI).
5. British National Formulary (BNF).